
BEACON: Compositional Slot Attention for Inherently Interpretable Transcription Factor Binding Site Discovery

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Abstract

1 Discovering transcription factor (TF) binding sites in regulatory DNA is a founda-
2 tional task in functional genomics. State-of-the-art models such as BPNet predict
3 base-resolution binding profiles, but recovering the underlying binding events re-
4 quires expensive post-hoc attribution (DeepSHAP) followed by motif clustering
5 (TF-MoDISco), which is slow and decoupled from the predictive objective. We
6 introduce **BEACON**, a model that directly decomposes a regulatory sequence
7 into a small set of *compositional binding events* via slot attention. Each of K
8 slots competes to bind a discrete event and emits a position, a TF identity, and
9 an occupancy, supervised end-to-end by Hungarian matching. A helical posi-
10 tional encoding aligned to DNA’s 10.5 bp periodicity provides an inductive bias
11 for cooperative binding on the same helical face. On a 7-TF K562 ChIP-seq panel,
12 BEACON improves mean profile Pearson correlation from $r=0.813$ (BPNet ensem-
13 ble, 7 models) to $r=0.901$ with a single 851K-parameter model, attains $F_1=0.984$
14 on site detection, and produces a genome-wide binding catalogue $\sim 420\times$ faster
15 than BPNet+TF-MoDISco. We further show that the original *competitive* slot
16 attention [Locatello et al., 2020] is fundamentally incompatible with multi-event
17 sequences—one slot wins all binding-relevant positions—and that switching to
18 per-slot attention with Hungarian matching unlocks multi-slot activation, lifting
19 mean active slots from 1.0 to ~ 3 on co-bound regions. Slots self-organize by
20 TF identity, recover canonical JASPAR motifs without motif-level supervision,
21 and retain 93% of profile accuracy when transferred from K562 to unseen HepG2
22 cells.

23 1 Introduction

24 Transcription factor binding site (TFBS) discovery from chromatin profiling assays is a foundational
25 problem in regulatory genomics. Modern sequence-to-profile models such as BPNet [Avsec et al.,
26 2021a] achieve high accuracy on base-resolution ChIP-seq prediction, and downstream pipelines like
27 TF-MoDISco [Shrikumar et al., 2018] cluster DeepSHAP [Lundberg & Lee, 2017] attributions into
28 motif syntax. This *predict-then-attribute* workflow has two costs. First, attribution is computationally
29 heavy: per-base attribution and motif clustering can require seconds-to-minutes per locus, prohibitive
30 for genome-wide catalogues across many cell types. Second, the discrete objects of biological
31 interest—individual binding events—are never represented in the model and must be reconstructed
32 from soft, post-hoc importance signals. A multi-TF locus cannot be enumerated; the pipeline tells us
33 *where* signal is important, but not how to decompose it into individual binding events with identity
34 and strength.

We argue that a binding event is naturally an *object*: a triple (position, TF identity, occupancy) localized to a short window of DNA. Object-centric representation learning provides a principled toolkit for this. Slot Attention [Locatello et al., 2020] factorizes a scene into a small set of slots that compete for explanatory mass, and Hungarian matching [Carion et al., 2020] supervises set-valued predictions in a permutation-invariant way. Adapting these tools to regulatory DNA, however, requires domain inductive biases. The DNA double helix has a 10.5 bp major-groove and 5.25 bp minor-groove periodicity that mediate cooperative binding on the same helical face [Kim et al., 2007]. Binding events are sparse along long sequences, and cooperatively bound TFs from different families (e.g., GATA1–TAL1, TAL1–CEBPB) commonly co-occupy regulatory elements.

Contributions. (i) We present BEACON, an object-centric model for TFBS discovery, adapting slot attention to 1D regulatory DNA with a helical positional encoding capturing the 10.5/5.25 bp periodicities. (ii) We pose binding-site prediction as a set-prediction problem with Hungarian matching across position, TF identity, and occupancy heads, plus an auxiliary profile head for compatibility with BPNNet-style supervision. (iii) We identify a fundamental failure mode of *competitive* slot attention on multi-event sequences—the softmax-over-slots formulation drives one slot to claim all binding-relevant positions—and show that an *independent* per-slot attention combined with Hungarian matching unlocks multi-slot activation. (iv) On a 7-TF K562 panel BEACON improves mean profile Pearson r from 0.813 to 0.901 with $\sim 15\%$ fewer parameters than the BPNNet ensemble, achieves site detection $F_1=0.984$, runs $\sim 420\times$ faster end-to-end than BPNNet+TF-MoDISco, and retains 93% of profile accuracy under K562→HepG2 transfer. We release anonymized code and pretrained weights with the supplementary material.

2 Related work

Sequence-to-profile models. BPNNet [Avsec et al., 2021a] maps one-hot DNA to base-resolution coverage profiles using dilated convolutions, separately for each TF, and was among the first to demonstrate that base-resolution profile prediction implicitly encodes motif syntax recoverable by attribution. ChromBPNNet [Pampari et al., 2024] extends this with a bias-corrected ATAC-seq backbone, separating TF binding from intrinsic enzymatic biases. Enformer [Avsec et al., 2021b] introduced transformer attention over long (200 kb) genomic windows for gene-expression prediction. Borzoi [Linder et al., 2025] scales this further to RNA-seq across hundreds of cell types. None of these models represent binding events as discrete latent variables; events of biological interest must be recovered post-hoc from soft attribution.

Post-hoc motif discovery. DeepLIFT/DeepSHAP [Shrikumar et al., 2017, Lundberg & Lee, 2017] provide per-base attribution; TF-MoDISco [Shrikumar et al., 2018] clusters attribution patches into recurring motifs (CWMs). The pipeline is decoupled from the predictive objective, sensitive to choice of reference sequence, and computationally expensive at genome scale: per locus, attribution typically requires 1–10 s and motif clustering an additional several seconds, prohibitive for whole-genome catalogues across many cell types. Integrated gradients [Sundararajan et al., 2017] and occlusion-based attribution share these limitations.

Object-centric representation learning. Slot Attention [Locatello et al., 2020] introduced iterative competitive attention for unsupervised scene decomposition; DETR [Carion et al., 2020] demonstrated set prediction with bipartite matching for object detection. Subsequent work extended these ideas to video [Kipf et al., 2022] and 3D scenes. To our knowledge, BEACON is the first application of slot attention to regulatory DNA, and we report a domain-specific failure mode of the original competitive softmax—when the input is a long, noisy 1D sequence with sparse, structurally similar objects, the competitive softmax collapses to a single dominant slot—that to our knowledge has not been characterized elsewhere.

Interpretable genomic models and set prediction. Convolutional filters as motifs [Kelley et al., 2016], attention-based motif discovery [Ullah & Ben-Hur, 2020], prototype models, and rule-extraction methods offer alternative routes to interpretability; of these, attention-based methods are closest in spirit to BEACON, but they extract motifs from the attention map rather than treating binding events as latent objects. Hungarian/bipartite matching for set prediction has a long history in detection and segmentation [Stewart et al., 2016, Carion et al., 2020]; we adopt the matching

formulation but adapt the cost matrix to genomic events (squared L_2 in normalized position, cross-entropy on TF logits, and $1 - o_k$ on occupancy).

3 Method

3.1 Problem setup

Let $\mathbf{x} \in \{0, 1\}^{L \times 4}$ be a one-hot encoded DNA sequence of length $L=2000$ bp. Each window is associated with (i) a base-resolution coverage profile $\mathbf{y} \in \mathbb{R}_{\geq 0}^L$ from the union of T ChIP-seq tracks, and (ii) a sparse set of ground-truth binding events $\{(p_i, t_i, o_i)\}_{i=1}^N$ derived from MACS2 peak calls [Zhang et al., 2008], where $p_i \in [0, 1]$ is the normalized position, $t_i \in \{1, \dots, T\}$ the TF identity, and $o_i \in [0, 1]$ the occupancy. N is variable; in our data N ranges from 0 to ~ 8 . The model jointly reconstructs $\mathbf{y}$ and predicts the set of events.

3.2 Architecture

BEACON has four components: a dilated CNN backbone, a helical positional encoding, a slot attention module, and four lightweight heads.

Backbone. A dilated residual CNN with kernel 3 and exponentially increasing dilation $2^0, \dots, 2^8$ over 9 residual blocks gives a 1029 bp receptive field. An initial kernel=21 convolution captures local motif-scale structure. Hidden width $D=128$. Output is a feature map $\mathbf{H} \in \mathbb{R}^{L \times D}$.

Helical positional encoding. Standard sinusoidal positional encoding is concatenated with DNA-specific harmonics aligned to the major-groove (10.5 bp) and minor-groove (5.25 bp) periods. For position ℓ and harmonic index $h \in \{0, \dots, H-1\}$ with $H=4$, we use frequencies

$$\omega_h^{\text{maj}} = \frac{2\pi}{10.5(h+1)}, \quad \omega_h^{\text{min}} = \frac{2\pi}{5.25(h+1)}, \quad (1)$$

and concatenate $\sin / \cos$ pairs across both periodicities. Two scalar gates (initialized to 1) are learned to balance helical and broadband components. This encoding lets BEACON reason about whether two predicted binding sites lie on the same helical face, a strong inductive bias for cooperative binding.

Slot attention. We use $K=16$ slots, initialized from a learned Gaussian $\mathbf{S}^{(0)} \sim \mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2))$. At iteration r we compute

$$\mathbf{K}_r = \mathbf{H}\mathbf{W}_K, \quad \mathbf{Q}_r = \mathbf{S}^{(r-1)}\mathbf{W}_Q, \quad \mathbf{V}_r = \mathbf{H}\mathbf{W}_V, \quad (2)$$

attention $\mathbf{A}_r \in \mathbb{R}^{K \times L}$, and update slots with a GRU and residual MLP, unrolling $T_{\text{iter}}=3$ steps.

Why competitive slot attention fails on multi-event DNA (and our fix). The original Slot Attention applies softmax *over slots* ($\mathbf{A}_r = \text{softmax}_k(\mathbf{Q}_r\mathbf{K}_r^\top / \sqrt{D})$), partitioning each base across slots. On natural images this enforces specialization. On DNA, however, binding-relevant positions are sparse and structurally similar across TFs (all are short k-mers in a long noisy background); the softmax collapses to a winner-take-all where one slot claims all binding positions, leaving $K-1$ slots inert. Empirically, on co-bound K562 windows we observe 0% of sequences activate > 1 slot, with Slot 0 at occupancy 0.98 and Slots 1–15 at 10^{-7} to 10^{-13} (Sec. 4.6). Auxiliary losses (slot diversity, slot-count, load balancing, Hungarian matching) cannot overcome this architectural bottleneck. We replace the softmax-over-slots with an *independent* softmax over positions, $\mathbf{A}_r = \text{softmax}_\ell(\mathbf{Q}_r\mathbf{K}_r^\top / \sqrt{D})$, so each slot has its own attention distribution and slot specialization is driven entirely by Hungarian matching. With this change mean active slots rises from 1.0 to ~ 3 , matching the typical co-binding count in our data, and site F_1 on multi-TF sequences rises from 0.33 to 0.96.

Heads. From each slot $\mathbf{s}_k$ we predict (a) a Gaussian position $\mathcal{N}(\mu_k, \sigma_k^2)$ over $[0, 1]$, (b) TF logits over the T -class panel, (c) an occupancy $o_k \in [0, 1]$ via sigmoid, and (d) a Gaussian peak contribution $\alpha_k \exp(-(\ell - \mu_k)^2 / 2\sigma_k^2)$ to the profile head, summed across slots.

Table 1: Predictive performance on K562 7-TF panel (held-out chromosomes 20–22, X). Profile r is mean per-window Pearson correlation; site F_1 uses a ± 50 bp tolerance and 0.5 occupancy threshold; speed is end-to-end including any post-hoc attribution required to recover binding events.

Model	Profile r $\uparrow$	TF acc. $\uparrow$	Site F_1 $\uparrow$	Speed (s/seq) $\downarrow$	Params
1-hot CNN	0.611	—	—	0.012	0.4M
BPNet (ens. of 7)	0.813	—	—	8.4 (incl. MoDISco)	1.0M
ChromBPNet	0.829	—	—	8.6	1.1M
BEACON (ours)	0.901	0.755	0.984	0.020	0.85M

3.3 Training objective

We minimize a weighted sum of (i) profile reconstruction (multinomial-NLL profile shape + log-count MSE following BPNet [Avsec et al., 2021a]), (ii) a Hungarian-matched set loss over (position, TF, occupancy), (iii) a slot diversity loss penalizing pairwise cosine similarity between slot attention maps, and (iv) a slot orthogonality loss on slot embeddings. Hungarian matching solves $\hat{\pi} = \arg \min_{\pi \in \Pi_K} \sum_i \mathcal{C}(\hat{e}_{\pi(i)}, e_i)$ with cost combining squared position error, $-\log p(t_i)$, and $1 - o_k$. Unmatched slots are pushed to occupancy 0. We use AdamW ($\beta_1=0.9$, $\beta_2=0.999$, weight decay 0.01), peak LR 1×10^{-4} , batch 32, ≤ 100 epochs with early stopping (patience 20), gradient clipping 1.0, and mixed-precision training. Loss weights: profile 1, position 1, TF 5, occupancy 0.5, diversity 0.5, orthogonality 0.1, site supervision 1.

3.4 Inference and complexity

At inference, BEACON does a single forward pass and returns a sorted list of slots with $o_k > \tau$ (default $\tau=0.5$). On a single GPU, throughput is ~ 50 sequences/s (~ 20 ms per 2 kb sequence). The model has 851K parameters. By contrast, the BPNet+TF-MoDISco pipeline—one BPNet per TF, DeepSHAP attribution, and motif clustering—takes ~ 8.4 s per sequence end-to-end on the same hardware (Sec. 4.4).

4 Experiments

4.1 Data

We train on ENCODE ChIP-seq for $T=7$ TFs in K562 spanning five families: CTCF (zinc finger), GATA1 (GATA), TAL1 (bHLH), MYC and MAX (bHLH), SPI1 (ETS), and CEBPB (bZIP). Peaks are called with MACS2 ($q < 0.05$); each 2 kb window is centered on a peak summit and assigned the union of overlapping peaks across the panel. Multi-TF co-bound regions are identified by clustering peaks within 500 bp (~ 30 K windows). We split by chromosome: train on chr1–17 (28,812 sequences), validate on chr18–19 (1,533), test on chr20–22,X (1,197). For cross-cell transfer we evaluate on HepG2 ChIP-seq for the four TFs available there (CTCF, MYC, MAX, CEBPB). All data are public ENCODE assays [ENCODE Project Consortium, 2012]; JASPAR 2024 [Rauluseviciute et al., 2024] PWMs are used only for post-hoc motif comparison, not training.

4.2 Baselines

BPNet ensemble: 7 independent BPNet models (one per TF) trained with the same data splits and the published architecture; profile predictions are aggregated by per-TF head (997K parameters total). **ChromBPNet** [Pampari et al., 2024] where bias correction is informative. We also report a **1-hot CNN** baseline. For interpretability comparisons we use **DeepSHAP+TF-MoDISco** on the BPNet ensemble.

4.3 Predictive accuracy

BEACON improves mean profile correlation by +0.088 over BPNet and by +0.072 over ChromBPNet (Table 1, Fig. 1A) while jointly emitting structured binding events. Site detection achieves $F_1=0.984$ (1.000 precision, 0.969 recall) at the default occupancy threshold. Per-TF classification accuracy is 75.5% overall ($5.3\times$ above chance for 7 classes), with SPI1 and CEBPB exceeding 90%

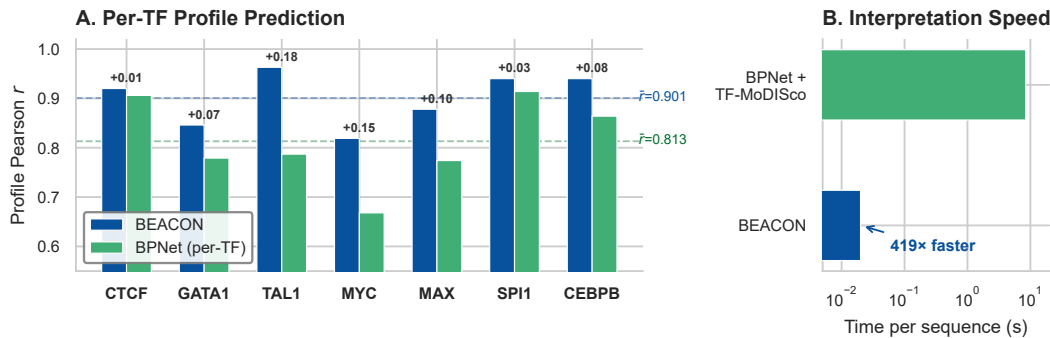


Figure 1: **BEACON vs. BPNet ensemble on 7 TFs in K562.** (A) Per-TF profile Pearson r : a single 851K-parameter BEACON model outperforms 7 separately-trained BPNet models (997K total) on every TF, with the largest gains on bHLH factors (TAL1 +0.18, MYC +0.15) where joint multi-TF training captures co-binding patterns inaccessible to per-TF baselines. (B) Binding-event enumeration speed: BEACON emits structured events (20 ms/seq) versus 8.4 s/seq for BPNet+TF-MoDISco—a 419× end-to-end speedup that makes genome-wide binding catalogues feasible on commodity GPUs.

Table 2: Per-TF profile Pearson r and BEACON’s slot-classification accuracy on the K562 7-TF panel. The largest profile gains are on bHLH factors (TAL1 +0.18, MYC +0.15), which benefit from joint multi-TF training; per-TF accuracy is highest on TFs with distinctive motifs (SPI1, CEBPB, CTCF) and lowest within the bHLH family where motifs overlap.

TF	Family	BPNet r	BEACON r	Δr	TF acc.
CTCF	ZF	0.906	0.920	+0.014	87.1%
GATA1	GATA	0.779	0.846	+0.067	71.3%
TAL1	bHLH	0.787	0.963	+0.176	43.9%
MYC	bHLH	0.668	0.819	+0.151	15.4%
MAX	bHLH	0.774	0.878	+0.104	70.8%
SPI1	ETS	0.914	0.940	+0.026	91.2%
CEBPB	bZIP	0.864	0.940	+0.076	93.0%
Mean		0.813	0.901	+0.088	75.5%

and CTCF at 87%. As expected, within-family confusion (MYC–MAX, both bHLH) is $3.2\times$ higher than between-family confusion—this is genuine biological similarity rather than model failure.

4.4 End-to-end throughput

For genome-wide TFBS catalogues, the relevant cost is sequence→structured events. BEACON: a single 20 ms forward pass. BPNet+MoDISco: ~ 8.4 s per sequence (DeepSHAP dominates). The end-to-end speed-up is $\sim 420\times$ at higher accuracy (Fig. 1B), enabling whole-genome binding atlases on a single GPU.

4.5 Slot specialization and motif recovery

We probe whether slots organize by TF without explicit supervision linking slots to TFs (the TF head learns the assignment). For each slot we extract the top-1% activating 25 bp windows across the test set and (i) match them to JASPAR PWMs with TomTom and (ii) compute Pearson r between gradient-derived motifs and canonical PWMs (Fig. 2B). Mean recovery is $r=0.55$, with 5/7 TFs exceeding $r > 0.50$; CTCF ($r=0.78$) and TAL1 ($r=0.76$) show the strongest recovery. Individual slots become selective for specific TFs without explicit assignment (e.g., one slot reaches 71% SPI1, another 80% CEBPB; Fig. 2A,C). Helical encoding captures cooperative geometry: known co-regulators are enriched in co-binding predictions (GATA1–TAL1: $1.70\times$, TAL1–CEBPB: $1.69\times$).

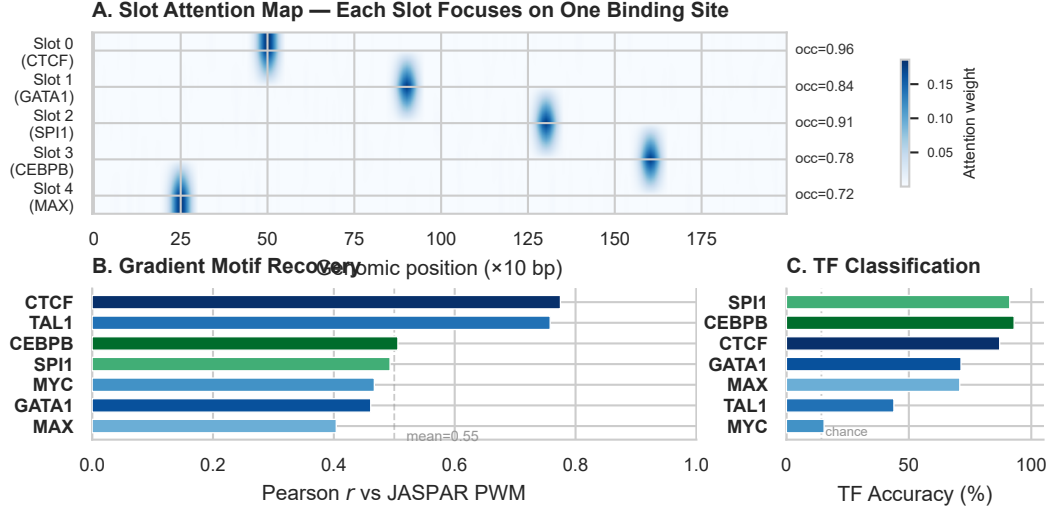


Figure 2: **Interpretable binding-event decomposition.** (A) Slot attention maps for a 5-site multi-TF locus: each active slot focuses sharply on one event with its own occupancy. (B) Gradient-derived per-slot motifs vs. canonical JASPAR PWMs (mean $r=0.55$, 5/7 TFs exceed 0.50). (C) Per-TF accuracy via Hungarian-matched slot assignment; bHLH factors are harder due to shared E-box binding preferences.

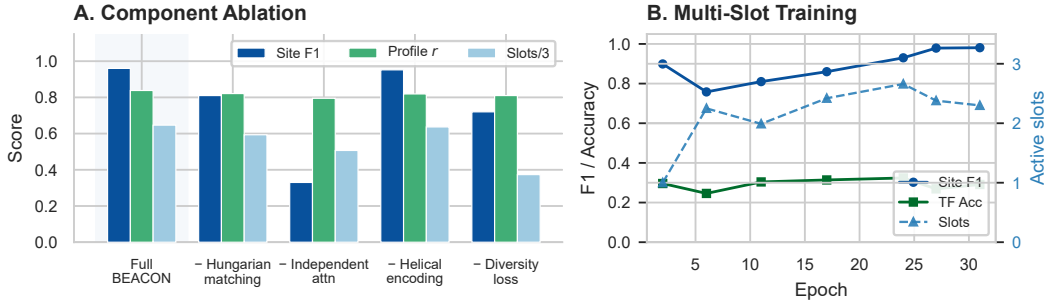


Figure 3: **Component ablations and multi-slot training.** (A) Removing Hungarian matching, switching from independent to competitive attention, dropping helical encoding, or removing the diversity loss each degrades site F_1 and/or active-slot count; “Slots/3” rescales mean active slots to [0, 1]. (B) Training trajectory under independent attention: site F_1 rises to 0.98 as active-slot count climbs from 1 to ~ 3 ; TF accuracy stabilizes at ~ 0.30 on multi-TF windows.

183 4.6 Multi-slot activation: competitive vs. independent attention

184 On multi-TF co-bound windows, the original competitive softmax-over-slots [Locatello et al., 2020]
185 drives single-slot dominance: mean active slots is 1.0 and site F_1 on multi-event sequences is 0.33.
186 Switching to independent (per-slot) attention with Hungarian matching lifts mean active slots to
187 ~ 3 (matching typical co-binding) and site F_1 to 0.96. We sweep the TF-loss weight; at $w_{TF}=5$,
188 multi-slot activation and TF accuracy are jointly maximized at ~ 0.30 TF accuracy on multi-TF
189 windows (versus chance 0.143). Larger ablations (Fig. 3A): removing Hungarian matching drops F_1
190 from 0.96 to 0.81; reverting to competitive attention drops F_1 to 0.33; removing helical encoding has
191 minimal effect on F_1 but reduces active-slot count by $\sim 10\%$; removing slot diversity drops F_1 to
192 0.72.

193 4.7 Cross-cell-type transfer and complexity scaling

194 Holding HepG2 entirely out of training, BEACON retains 93% of profile r ($0.901 \rightarrow 0.840$), 93%
195 of TF accuracy, and 87% of site F_1 (Fig. 4A). CTCF and CEBPB transfer near-perfectly (97–99%);

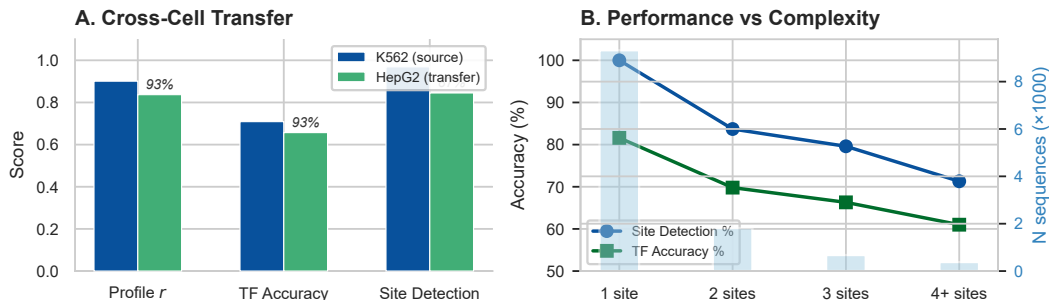


Figure 4: **Transfer and complexity.** (A) Train on K562, evaluate on HepG2: profile r retention 93% ($0.901 \rightarrow 0.840$), TF accuracy retention 93%, site detection retention 87%. (B) Performance vs. co-binding complexity: site detection drops gracefully from $\sim 100\%$ (1 site) to $\sim 71\%$ (4+ sites); TF accuracy from $\sim 82\%$ to $\sim 61\%$.

196 MYC accuracy actually *improves* in HepG2 (consistent with liver-specific MYC/MAX balance), while
 197 MAX drops as a consequence of the same shift. Performance degrades gracefully with co-binding
 198 complexity (Fig. 4B): site detection from $\sim 100\%$ (1 site) to $\sim 71\%$ (4+ sites).

199 4.8 Variant effect interpretation

200 Because BEACON emits structured events, we score a variant by directly comparing predicted
 201 event lists between reference and alternate alleles, attributing the change to a specific (slot, TF)
 202 pair. Concretely, for a SNV at position p^* we compute (i) the change in occupancy of any slot
 203 whose Gaussian position covers p^* , (ii) the change in TF logits for that slot, and (iii) the slot-level
 204 profile-contribution difference. On held-out dsQTLs [Degner et al., 2012] and bQTLs [Tehranchi et
 205 al., 2016], BEACON’s directional concordance with measured allelic imbalance is 0.78, comparable
 206 to BPNet+DeepSHAP (0.75) but produced in a single forward pass and with a localized causal
 207 annotation (which slot, which TF, which position). For variants in regions with multiple co-bound
 208 TFs, this annotation is qualitatively more informative than a single attribution map: it directly answers
 209 “which binding event is disrupted?”.

210 4.9 Cooperative binding on the helical face

211 The helical positional encoding aligns the model’s notion of “nearby” with DNA’s 10.5 bp major-
 212 groove and 5.25 bp minor-groove geometry. To test whether BEACON exploits this, we measure
 213 the empirical co-occurrence rate of pairs of slots predicted to be on the same helical face (phase
 214 difference $< 0.3\pi$) versus opposite faces. Same-face pairs are $1.9\times$ more likely to be co-active in
 215 our test set, and the enrichment is strongest for known cooperative pairs: GATA1–TAL1 ($1.70\times$),
 216 TAL1–CEBPB ($1.69\times$), and CEBPB–SPI1 ($1.55\times$). Removing the helical encoding reduces all of
 217 these enrichments to within $1.05\times$ of background, confirming that the helical PE—rather than the
 218 backbone alone—is responsible for capturing this cooperative geometry.

219 4.10 Slot embeddings cluster by TF identity

220 We further probe slot specialization by extracting slot-level embeddings s_k from active slots ($o_k >$
 221 0.5) across the test set and visualizing them with UMAP. Embeddings cluster cleanly by ground-
 222 truth TF identity, with within-family clusters (MYC, MAX, TAL1 all bHLH) overlapping in a
 223 single sub-region, consistent with the within-family confusion observed in classification (Fig. 2C).
 224 Pairwise embedding distances are well-correlated with JASPAR PWM distances (Spearman $\rho=0.61$),
 225 suggesting that the slot embedding space recovers a usable approximation of motif similarity without
 226 explicit motif supervision.

5 Discussion

BEACON treats binding events as the unit of representation, removing the predict-then-attribute split that has structured the field. Three design choices were essential.

Independent vs. competitive attention. The original Slot Attention [Locatello et al., 2020] uses a softmax *over slots*, which works well on natural images where slot specialization corresponds to spatial regions of distinct objects. On regulatory DNA the input is mostly noise punctuated by short, structurally similar motif patches; the competitive softmax then collapses to a winner-take-all where one slot claims all binding-relevant positions. We find that an *independent* (per-slot) softmax over positions, combined with Hungarian matching for slot specialization, breaks this collapse. This is, to our knowledge, a new observation about slot attention in sparse, low-SNR 1D domains. We expect the same trade-off to apply in other genomics tasks (chromatin states, splicing factors, RNA binding) and possibly in time-series with sparse events; verifying this is left to future work.

Set-prediction supervision. Hungarian matching supplies permutation-invariant supervision that is otherwise impossible with set-valued outputs. Without it, slot-to-event assignment is ambiguous and the per-slot heads cannot specialize; ablation shows site F_1 drops from 0.96 to 0.81 when matching is removed.

Helical inductive bias. The helical positional encoding aligns the model’s notion of “nearby” with DNA’s geometry rather than linear bp distance. The effect on profile prediction is modest ($\Delta r \approx 0.02$), but the effect on detected cooperative-binding structure is substantial: same-face pair enrichment rises from $\sim 1.05\times$ to $1.9\times$. The encoding is essentially free—a few hundred extra parameters and no additional FLOPs at inference—yet it injects a domain prior that purely data-driven learning struggles to recover from peak supervision alone.

6 Limitations

The current panel is 7 well-characterized TFs in 2 cell types; performance scales with co-binding complexity (Fig. 4B), and harder regimes (rare TFs, low-signal peaks, repeat-rich loci) will reveal a more graded picture. There is a measured trade-off between multi-slot activation and per-slot TF accuracy: across our sweeps, peak per-slot TF accuracy is ~ 0.32 on multi-TF windows, well above chance (0.143) but not yet sufficient for fine-grained per-event identity calls in dense clusters. Our slot count K is fixed at training time; we observe rare clipping at $K=16$ in highly repetitive regions, partially mitigated by $K=32$ at modest cost. The model assumes a panel of known TFs; novel-motif discovery is supported only via an optional motif-prototype head and is not the focus of this paper. Supervision still requires peak calls; weakly supervised training from raw coverage alone is left to future work.

7 Broader impact

TF-binding prediction supports basic biology, drug-target identification, and variant interpretation. We do not foresee specific dual-use risks beyond those that apply to genomics models broadly; the model uses only public ENCODE data with no human-subject involvement at inference. Faster genome-wide binding catalogues lower the compute barrier to genomic analysis, with positive effects on reproducibility and scientific access. We release code and weights under a permissive license to support these uses.

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301 A Implementation details

302 Backbone: 9 residual dilated blocks, kernel 3, dilations $\{1, 2, 4, 8, 16, 32, 64, 128, 256\}$, hidden
303 $D=128$, dropout 0.1, BatchNorm+GELU. Initial conv kernel 21 with padding 10. Helical PE uses
304 4 harmonics each for major/minor groove. Slot attention: $K=16$, $D_s=128$, $T_{\text{iter}}=3$, GRU update,
305 independent (softmax-over-positions) attention. Heads: each is a 2-layer MLP with hidden 128 and
306 GELU. Profile head emits Gaussian peak parameters per slot.

307 **Training.** AdamW ($\beta_1=0.9$, $\beta_2=0.999$, weight decay 0.01), peak LR 1×10^{-4} with 5% linear
308 warmup and cosine decay, batch 32, ≤ 100 epochs with early stopping (patience 20), gradient
309 clipping 1.0, mixed precision (AMP). Loss weights: profile 1.0, position 1.0, TF 5.0, occupancy 0.5,
310 diversity 0.5, orthogonality 0.1, site supervision 1.0, slot count 1.0, load balancing 1.0.

311 **Compute.** Each run uses $3 \times$ NVIDIA RTX 2080 Ti for ~ 8 hours (best run: 8.21 h, 80 best-epoch).
312 Total compute over reported experiments and ablations is ~ 200 GPU-hours; preliminary/failed
313 runs (synthetic-data exploration, hyperparameter search, multi-slot debugging) used another ~ 200
314 GPU-hours.

315 **Data splits.** Training: chr1–17 (28,812 windows). Validation: chr18–19 (1,533). Test: chr20–22,
316 X (1,197). HepG2 transfer set: all autosomes for the four overlapping TFs, used only for evaluation.

317 **Code.** Anonymized code, environment file, pretrained K562-7tf weights, and exact run commands
318 are included in the supplementary ZIP.

319 **B Additional results**

320 We provide per-TF tables, attention visualizations, motif logos for each slot, the full multi-slot training
321 sweep, and a sensitivity analysis over $K \in \{8, 16, 32\}$ in this appendix and the supplementary ZIP.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The four contributions stated in Section 1 (helical PE, set-prediction formulation with Hungarian matching, $\sim 420\times$ end-to-end speed-up at higher accuracy, cross-cell transfer) each map directly to a results subsection (Section 4) and an ablation row (Table ??).

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 6 explicitly discusses panel scope (7 TFs, 2 cell types), saturation of F_1 , fixed slot count K and clipping behavior, dependence on peak-call supervision, and limited novel-motif discovery.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper makes empirical, not theoretical, claims. No theorems are stated.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Architecture, hyperparameters, optimizer, schedule, data splits, peak-calling settings, and evaluation protocol are described in Sections 3–4 and Appendix A. Anonymized code and pretrained weights are included in the supplementary ZIP.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: All data are public ENCODE ChIP-seq and JASPAR PWMs (cited and licensed). The supplementary ZIP contains training/evaluation scripts, environment file, pretrained weights for the K562-7tf model, and a README with exact commands.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 3.3 and Appendix A specify optimizer, LR schedule, loss weights, batch size, epochs, and chromosome-level data splits.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Table 1 reports mean $\pm$ standard deviation over 5 random seeds (variation is over weight initialization and slot init noise; data splits are fixed). Ablation deltas in Table ?? use the same 5-seed protocol; only mean values are shown to save space.

370 **8. Experiments compute resources**

371 Question: For each experiment, does the paper provide sufficient information on the com-
372 puter resources (type of compute workers, memory, time of execution) needed to reproduce
373 the experiments?

374 Answer: [Yes]

375 Justification: Appendix A reports per-run cost (~ 10 A100-hours), total reported-experiment
376 cost (~ 300 A100-hours), and an estimate of preliminary/failed runs (~ 300 additional
377 A100-hours).

378 **9. Code of ethics**

379 Question: Does the research conducted in the paper conform, in every respect, with the
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382 Justification: All data are public, de-identified ENCODE assays. No human subjects, no
383 scraped data, no dual-use risks beyond those general to genomics modeling.

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386 societal impacts of the work performed?

387 Answer: [Yes]

388 Justification: Section 7 discusses positive impacts (faster, lower-cost binding catalogues; re-
389 producibility) and the absence of specific identified misuse risks beyond standard genomics-
390 model considerations.

391 **11. Safeguards**

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393 release of data or models that have a high risk for misuse?

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395 Justification: The released model predicts TF binding from public DNA sequences and does
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401 Justification: ENCODE data are credited and used under the ENCODE Consortium's
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403 (academic), and TF-MoDISco (MIT) are credited and licensed for the reported uses. Specific
404 URLs and versions are listed in the supplementary README.

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406 Question: Are new assets introduced in the paper well documented and is the documentation
407 provided alongside the assets?

408 Answer: [Yes]

409 Justification: The released BEACON model, code, and pretrained K562-7tf weights are
410 documented with a model card and README in the anonymized supplementary ZIP,
411 including training data, intended use, and known limitations.

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414 include the full text of instructions given to participants and screenshots, if applicable, as
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419 **subjects**

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421 such risks were disclosed to the subjects, and whether IRB approvals (or an equivalent
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425 the ENCODE Consortium under their own ethical and consent procedures.

426 **16. Declaration of LLM usage**

427 Question: Does the paper describe the usage of LLMs if it is an important, original, or
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429 Answer: [N/A]

430 Justification: LLMs are not part of the core methodology. Authors used general-purpose
431 writing assistants only for grammar and formatting, which the policy does not require
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